Han Yang

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Education

University of Massachusetts Amherst — Sep. 2025 - Present
Ph.D. in Computer Science — *Amherst, MA*

- Advisor: Prof. Chuang Gan
- Research Interest: Robotics, Spatial Understanding, Embodied AI

University of Pennsylvania — Aug. 2023 - Dec. 2023
Exchange in School of Engineering and Applied Science — *Philadelphia, PA*

- GPA: **4.000**/ 4.000
- Advanced Courses: Computer Graphics; Natural Language Processing; Applied Machine Learning

The Chinese University of Hong Kong — Sep. 2021 - May. 2025
B.Sc. in Computer Science — *Hong Kong SAR, China*

- GPA: **3.803**/ 4.000 (top 5%)
- Advisor: Prof. Chi-Wing Fu
- Research Interest: Computer Vision, Multimodal Learning

Publications & Manuscripts

- **PhyScensis: Physics-Augmented LLM Agents for Complex Physical Scene Generation** — **ICLR 2026** (Under Review)
  Yian Wang*, **Han Yang***, Minghao Guo, Xiaowen Qiu, Tsun-Hsuan Wang, Wojciech Matusik, Joshua B. Tenenbaum, and Chuang Gan.
  Introduced an agentic framework that leverages procedural rules and a physics simulator to generate physically plausible and complex 3D scenes.

- **3D-Mem: 3D Scene Memory for Embodied Exploration and Reasoning** — **CVPR 2025**
  Yuncong Yang*, **Han Yang***, Jiachen Zhou, Peihao Chen, Hongxin Zhang, Yilun Du, and Chuang Gan.
  Introduced 3D-Mem, a 3D scene memory framework that empowers VLM agents with lifelong memory, exploration, and reasoning abilities in 3D environments.

- **UniMuMo: Unified Text, Music, and Motion Generation** — **AAAI 2025 Oral**
  Han Yang, Kun Su, Yutong Zhang, Jiaben Chen, Kaizhi Qian, Gaowen Liu, and Chuang Gan.
  Developed UniMuMo, a unified generative model that seamlessly performs cross-modal generation between text, music, and human motion within a single, shared framework.

- **SILT: A Shadow-aware Iterative Label Tuning Approach for Learning to Detect Shadows from Noisy Labels** — **ICCV 2023**
  Han Yang*, Tianyu Wang*, Xiaowei Hu, and Chi-Wing Fu.
  Proposed SILT, a novel iterative label tuning approach that leverages shadow-aware physical properties to progressively refine noisy annotations and train robust shadow detection models.

Experience

- **University of Massachusetts Amherst**
  Doctoral Research Assistant, advised by Prof. Chuang Gan
  Embodied AI, Robotics, Large Language Model

- **MIT-IBM Watson AI Lab**
  Research Assistant Intern, advised by Prof. Yilun Du and Prof. Chuang Gan
  Audio Processing, Diffusion Model, Multimodal Learning, Embodied AI

- **Division of Visual and Interactive Computing (VIC) Lab**
  Research Assistant Intern, advised by Prof. Chi-Wing Fu
  Self-supervised Learning, Unsupervised Learning, Low-level Vision

Awards & Honors

Yasumoto International Exchange Scholarship (2023, top 1 in ongoing exchange students)

Arthur and Louise May Memorial Scholarship (2023, top 1 in CS major);

Professor Omar Wing Memorial Scholarship (2022, top 1 in CS major)

Award for Outstanding Academic Performance (2023-2025, top 5% in CS major)

Dean's List (2022-2025, top 10% in the department)

Skills

Languages	Mandarin (native), English (TOEFL 106, GRE 330)
Programming	Python, PyTorch, C/C++, OpenGL, Java, html, LaTeX
Others	Figma, Pr, SolidWorks, Blender, TeleportHQ